Do Options Overprice Macro News? SPY Options around CPI, Payrolls and FOMC Decisions, 2022–2026

The volatility risk premium on announcement days, its tradability, and when event risk leaves the price

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ABSTRACT

Options usually charge more for protection than the moves they protect against turn out to be: the volatility risk premium. This paper asks whether the premium survives on the days the market fears most, the US consumer price index (CPI), employment report (payrolls) and Federal Reserve (FOMC) decision days. It compares the next-day move priced by VIX1D with the S&P 500's actual move on 1,121 days from May 2022 to November 2026, trades real at-the-money SPY straddles expiring the next day on 267 days from October 2025, and follows the variance priced by those options minute by minute with a Black–Scholes pricer written for the purpose, which reproduces the data vendor's implied volatilities to within 3×10^{-5} volatility points. On normal days the S&P moves 0.89 times what options priced [95% CI 0.84, 0.94]. On release days options charge 27% more than on normal days but the index moves 43% more, so the premium shrinks; the difference holds on every robustness check but is not statistically significant overall. Payrolls is the event options miss: they price almost no extra overnight risk for it, yet the overnight move is 1.5 times what was priced. CPI options were too cheap in the 2022 inflation shock and too expensive since 2023. After trading costs, selling the straddle every day has a Sharpe ratio of 0.23 over 2022–26; skipping release days raises it to 0.52 and halves the worst drawdown. On FOMC days most of the priced risk leaves during the press conference, not at the 14:00 statement.

Keywords: volatility risk premium, macroeconomic announcements, options, VIX1D, implied volatility, event risk.
JEL: G13, G14, E44.

1. Introduction

Insurance is sold at a profit, and options are no exception. Across many markets and decades, the volatility implied by option prices has on average exceeded the volatility that followed: buyers pay a premium for protection against large moves, and sellers earn it as compensation for losing heavily when those moves come (Bakshi and Kapadia, 2003; Carr and Wu, 2009). A natural question is whether that premium is uniform over time or concentrated on particular days.

Scheduled macroeconomic announcements are the obvious candidates. The CPI and the employment report are released at 08:30 New York time before the stock market opens, and the Federal Reserve announces its decision at 14:00 followed by a press conference at 14:30. A large share of the equity premium is earned on such days (Savor and Wilson, 2013; Lucca and Moench, 2015), and option prices rise ahead of them in the same way they rise ahead of company earnings (Dubinsky, Johannes, Kaeck and Seeger, 2019). If investors overpay for protection on ordinary days, they might be expected to overpay even more on the days they fear most.

Two recent developments make the question answerable at a daily frequency. Since 2022 SPY and S&P 500 options expire every trading day, so an option bought on the evening before a release expires at the close of the release day and prices that day's risk alone. And since May 2022 Cboe has published VIX1D, the market's expected volatility for the next trading day. This paper uses both. It asks four questions: whether options overprice the next day's move on release days as they do on normal days; whether the answer rests on a handful of extreme days; whether the premium can be earned on real options after costs and at what risk; and when, within a release day, the event risk leaves option prices.

Main findings. (1) Options overprice normal days: the S&P moves 0.89 times what was priced, robustly. (2) On release days the premium shrinks; options charge 27% more than on normal days but the index moves 43% more. The direction holds on every check, the overall difference is not significant. (3) Payrolls is the event options miss, priced as almost no extra overnight risk while moving the index 1.5 times what was priced. (4) CPI options switched from too cheap in 2022 to too expensive since 2023. (5) After costs, skipping release days doubles the Sharpe ratio of a straddle seller and halves the drawdown. (6) For the Fed, the risk leaves during the press conference.

The rest of the paper describes the data and the trade (Section 2), the checks the data passed before any result was estimated (Section 3), the 4.5-year comparison (Section 4) and its robustness (Section 5), the real trade (Section 6), the pricer (Section 7) and the minute-by-minute evidence (Section 8).

2. Data and the trade

The trade studied is the at-the-money *straddle*: at the close of the evening before a day E (the *eve*), buy a call and a put on SPY with the strike nearest SPY's price, both expiring at the close of E. It pays |SPY at expiry − strike|, so its price is the absolute move the market charges for, and selling it earns the difference between that price and the move that happens. Since 2025 every trading day is an expiry day for SPY options.

VIX1D is quoted as an annualised volatility. For a normal distribution the expected absolute move is $\sqrt{(2/\pi)} \approx 0.80$ times the standard deviation, so the absolute move VIX1D implies for the next day is

$$\text{implied } |\text{move}| = 0.80 \times \text{VIX1D} / \sqrt{252}. \tag{1}$$

VIX1D is built from S&P 500 index options rather than SPY options and from the whole range of strikes rather than the at-the-money one, so it is not itself a straddle price. On the 267 eves on which both exist, the SPY straddle costs a median 0.94 times the move in (1), with a correlation of 0.971 (Figure 1). That factor turns VIX1D into a straddle price for the 4.5-year sample, which is then checked against the real trade in Section 6.

Table 1. Data.

Source	What	Period
Cboe	VIX1D, VIX9D, VIX, VIX3M, daily close	May 2022 – Nov 2026
FRED	S&P 500 close; 3-month Treasury bill rate	May 2022 – Nov 2026
London Strategic Edge	SPY 1-minute bars	May 2022 – Nov 2026
	SPY options expiring the next day, 3 strikes around the eve's close: 1-minute candles	Oct 2025 – Nov 2026
	Minute implied volatility and Greeks of the at-the-money pair, release days	Oct 2025 – Oct 2026
	Snapshot of the full SPY chain (expiries up to 44 days)	3 Nov 2026
	US macroeconomic calendar (CPI, payrolls, FOMC)	2022 – 2026
Public sources	SPY quarterly dividends (ex-dates and amounts)	2025 – 2026

Release days: 51 CPI, 50 payrolls and 35 FOMC days in the long sample; 11, 11 and 9 with real option prices. A day with two releases takes the label CPI, then payrolls, then FOMC. Half days (13:00 close) are excluded.

3. Validation: problems found and checks passed

Every number in this paper depends on prices being what they claim to be, so the data were checked before any result was estimated. Table 2 lists the problems found. Two deserve comment.

Table 2. Data problems found and fixed.

Problem	How it was found	Fix
SPY resolved to a paid Nasdaq feed	"feed requires subscription" errors	Request the free US stocks dataset by name
Option candles exist only from Oct 2025, though the catalogue lists trades since 2014	Size estimates month by month before downloading	Real options for 2025–26; VIX1D for 2022–26, checked against them
Put–call parity off by +$0.11, and by −$1.70 on four days	Parity check at the eve's close	Interest over the settlement gap and dividends (below)
Half days close at 13:00	Last bar before 15:59	Excluded; the pricer uses the 13:00 close
A "payrolls" day that was the tariff crash (4 Apr 2025)	Worst day for a seller	Kept; robustness rules apply equally to all days
No bid or ask quotes	Trades only	Costs as scenarios, anchored on expiry prices

Put–call parity. A call bought and a put sold on the same strike K are equivalent to owning SPY and owing K at expiry, so at the eve's close call − put should equal SPY − K plus the interest on K until it is paid, minus any dividend before expiry. Observed call − put exceeded SPY − K by a median $0.11 on 267 eves, with all prices taken in the same minute. The size of the interest term depends on when money moves. Trades settle one business day later (T+1): SPY bought at the eve's close is paid the next settlement day, while with the options K is paid when the exercise settles, one settlement day after expiry. The interest therefore covers one calendar day on most eves but three on Thursday eves, whose Friday exercise settles on Monday, and four or zero around Columbus and Veterans Day, when stocks trade but banks are closed. The data show exactly this pattern: the gap is $0.09 on one-day eves (predicted from the T-bill rate: $0.07) and $0.25 on Thursday eves (predicted $0.225). On the four days on which SPY went ex-dividend at expiry, the gap of about −$1.70 is the dividend; after the dividend and three days of interest, three of them are within 7 cents. With both terms accounted for, the median gap falls from 11 cents to 2 and 91% of eves are within 25 cents. The rate implied by the remaining gap, about 4.7% a year against a 3.7% T-bill, is consistent with market makers funding themselves above the government rate. A cluster of late-June 2026 eves remains off by $0.5–2; excluding those days changes no result (Section 6).

Other checks. Every eve price was traded in the 15:59 minute. At expiry, options last trade a median 1–2 cents above their intrinsic value, the price of immediacy that anchors the cost scenarios in Section 6. And VIX1D behaves as an event measure should: across a CPI day it falls from a median 17.4 to 14.1 and across a payrolls day from 17.5 to 14.6, while on normal days it barely moves (13.6 to 13.8). FOMC days are the exception, with VIX1D as high the evening after the decision as before it (17.1 and 17.0).

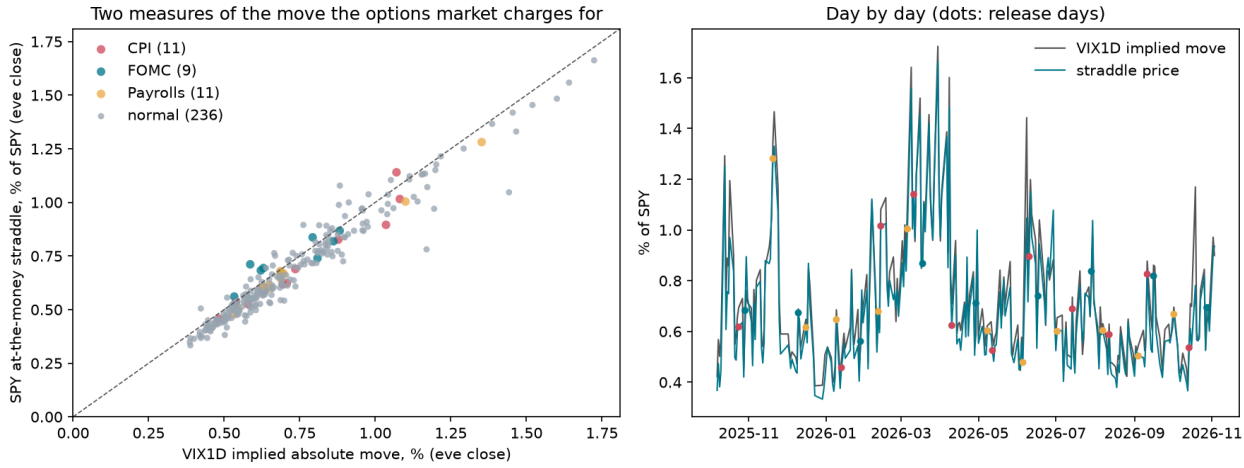


Figure 1. Left: price of the SPY at-the-money straddle against the absolute move implied by VIX1D, at the eve's close, 267 eves (October 2025 – November 2026). Right: the two series day by day; dots mark release days.

4. Options against outcomes, May 2022 – November 2026

For each of 1,121 trading days, the absolute move implied by VIX1D at the previous close is compared with the S&P 500's absolute log return over the day. Table 3 reports the ratio of the two totals and of the variances, with 95% bootstrap confidence intervals, and the profit of a straddle seller whose price is 0.94 times the implied move.

Table 3. Actual against implied moves, by kind of day (95% CI).

Day	Days	Implied |move|	Actual |move|	Actual / implied	Variance ratio	Seller per trade	Seller wins
CPI	51	1.04%	0.98%	0.94 [0.71, 1.21]	1.04 [0.52, 1.69]	−0.004%	59%
Payrolls	50	0.94%	1.05%	1.11 [0.89, 1.38]	1.35 [0.71, 2.28]	−0.162%	44%
FOMC	35	0.97%	0.91%	0.94 [0.71, 1.20]	0.83 [0.50, 1.30]	+0.002%	54%
Normal	985	0.78%	0.69%	**0.89 [0.84, 0.94]**	0.83 [0.69, 0.99]	+0.037%	62%

Seller profit before costs, in % of the index per trade. Releases minus normal days: −0.097% [−0.248, +0.046].

On normal days the result is the familiar one: the index moves 11% less than options priced, and a seller wins on 62% of days. On release days options are more expensive, a 27% higher implied move than on normal days on average, but the index moves 43% more. Their ratios are close to or above one and the seller earns nothing on average; on payrolls days the seller loses 0.16% per trade. The release-day estimates are imprecise, however: each rests on 35–51 days, and the difference between release and normal days in seller profit, −0.10%, has a confidence interval that includes zero. Figure 2 shows the cumulative profit of selling every evening: steady on normal days apart from the spring of 2025, flat or negative on release days.

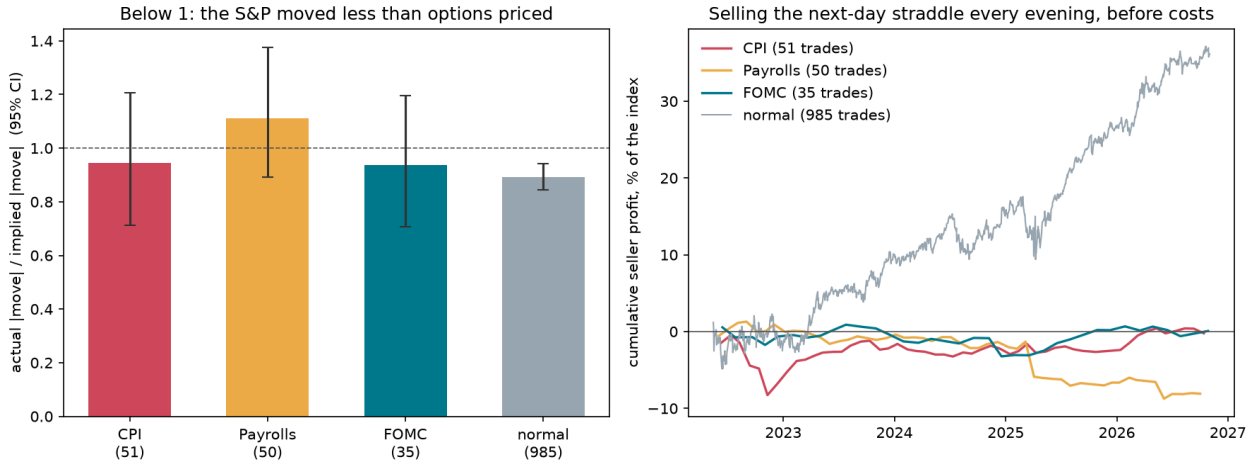


Figure 2. Left: actual over implied absolute move by kind of day (95% CI); below 1 the index moved less than options priced. Right: cumulative profit of selling the next-day straddle every evening, by kind of day, in % of the index, before costs.

5. Is it a handful of days?

Averages over 50 days can be driven by one or two. One of the worst payrolls days for a seller, 4 April 2025, was really the tariff crash: China announced 34% retaliatory tariffs that morning and the index fell 6.2%. Removing it by hand because it hurts the result would be cherry-picking. Instead each check in Table 4 applies the same rule to every kind of day.

Table 4. Actual over implied absolute move under five checks.

Check	CPI	Payrolls	FOMC	Normal
All days (Table 3)	0.94	1.11	0.94	0.89
Largest 5% of surprises removed from each group	0.79	0.95	0.84	0.79
Median day (fair under normality: 0.85)	0.80	0.99	0.79	0.73
Rank test against normal days, p-value	0.52	0.06	0.51	
Without the single most influential day	0.88	1.02	0.87	0.89
2022	1.48 [0.77, 2.26]	0.83	1.01	0.95
2023 – 2026	**0.78 [0.60, 0.96]**	1.17 [0.93, 1.48]	0.91	0.88 [0.83, 0.93]

Surprise = actual over implied absolute move on one day. Rank test: one-sided Mann–Whitney, release-day surprises larger than normal-day ones. All release days pooled: p = 0.19.

Three conclusions follow. First, normal days are overpriced on every measure, in both periods. Second, release days as a group are priced less generously than normal days on every measure: with the same trim the gap is about 0.08, and on medians options charge 26% more than on normal days while the index moves 44% more, almost exactly the averages of Section 4. But a rank test cannot distinguish release-day surprises from normal-day ones (p = 0.19), so the shrinking premium is a consistent pattern rather than an established fact. Third, the kinds of release differ. Payrolls stay above normal days under every rule and are borderline significant (p = 0.06); without the tariff day the ratio falls from 1.11 to 1.02, still above normal days' 0.89. CPI and FOMC days, once their largest surprises are trimmed, look like normal days.

CPI has two regimes. In 2022, when inflation peaked at 9.1%, CPI days moved 1.48 times their implied move and a seller lost about 1% of the index per CPI day. Since 2023 they have moved 0.78 times, a confidence interval below one: CPI options have been too expensive. Meanwhile the premium options charge on CPI days over normal days fell from 1.54 times in 2022 to 1.08 times in 2026, as if the market slowly learned that inflation had stopped surprising. The split was chosen after seeing the year-by-year numbers, so this is a hypothesis for 2027 data to test, not a result.

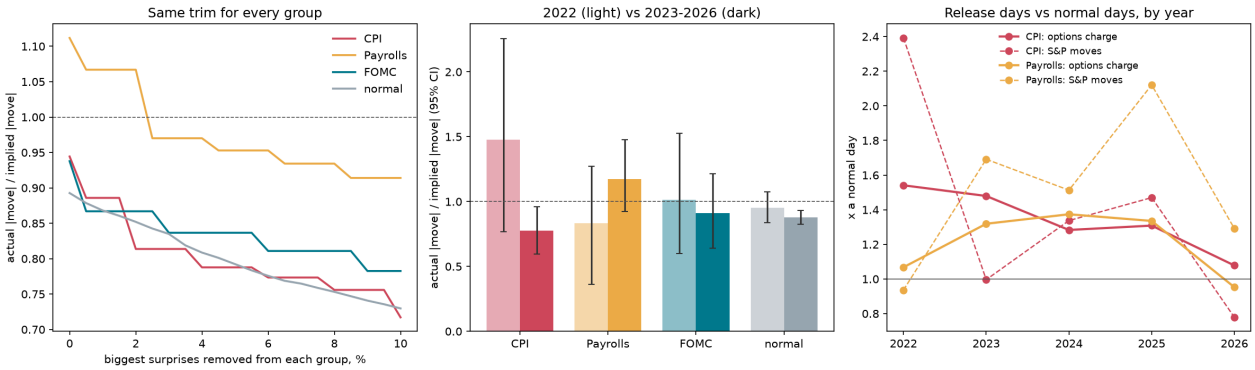


Figure 3. Left: actual over implied move as the same share of each group's largest surprises is removed. Centre: 2022 against 2023–2026 (95% CI). Right: how much more options charge on CPI and payrolls days than on normal days, and how much more the index moves, year by year.

6. The real trade

For the year in which real prices exist, the straddle is sold at the eve's last trade (15:59) and settled at |SPY close − strike|. On the 266 days covered by both, the VIX1D-based profit of Sections 4–5 and the real profit have a correlation of 0.989 and a typical gap of 0.05% of the index, so the long sample is a faithful proxy. The exception is FOMC days, where the real straddle costs 1.05 times the VIX1D move instead of 0.94 and earns more (+0.094% against +0.050%).

Trading costs are modelled as scenarios because the data contain trades but not quotes. A professional is assumed to pay a 1-cent half-spread and $0.20 per contract per option traded, a retail trader 2 cents and $0.70. Selling a straddle trades three options: the call and the put on the eve, and the one that ends in the money bought back before expiry.

Table 5. Real SPY straddles, seller profit per trade, October 2025 – November 2026.

Day	Days	Before costs [95% CI]	Professional	Retail	Wins	$ per straddle (retail)
CPI	11	+0.218% [−0.079, +0.505]	+0.213%	+0.206%	64%	+138
Payrolls	11	−0.124% [−0.592, +0.219]	−0.129%	−0.135%	45%	−103
FOMC	9	+0.094% [−0.248, +0.408]	+0.089%	+0.083%	67%	+59
Normal	236	+0.051% [−0.014, +0.114]	+0.046%	+0.040%	59%	+28

Median premium $4.41 a share ($441 a straddle), 0.60% of SPY. Retail costs: $0.08 a share, 0.011% of SPY per straddle. A straddle is 100 shares.

Costs take about a fifth of the normal-day edge for a retail trader and less for a professional: the premium survives them. The pattern across release days repeats Section 5 on new data, with CPI options expensive and payrolls options cheap. Table 6 turns the trade into strategies, after retail costs. Over 2022–26, selling every evening earns a Sharpe ratio of 0.23 with a worst drawdown of 16% of the index; skipping release days raises the Sharpe ratio to 0.52 and halves the drawdown. The real year flatters every strategy: it was calm, and there skipping release days did not help, because CPI options happened to be the most overpriced of all. The tail risk is the price of the premium. The worst day for a seller, 9 April 2025, when a pause in tariffs lifted the index 9.1%, cost 5.6% of the index, about eight days of premium.

Table 6. Strategies after retail costs.

Strategy	Trades	Mean	Sharpe (no costs)	Worst day	Worst 5% of trades	Worst drawdown
VIX1D proxy, May 2022 – November 2026						
Sell every day	1,121	+0.009%	0.23 (0.61)	−5.63%	−1.69%	−16.3%
Skip release days	985	+0.021%	**0.52** (0.89)	−5.63%	−1.55%	**−8.6%**
Release days only	136	−0.076%	−0.49 (−0.39)	−4.58%	−2.64%	−13.9%
Real straddles, October 2025 – November 2026						
Sell every day	267	+0.041%	1.28 (1.63)	−2.23%	−1.29%	−3.5%
Skip release days	236	+0.040%	1.20 (1.54)	−2.23%	−1.26%	−2.8%

Profit in % of the index per trade. Sharpe ratio annualised over all trading days, counting days without a trade as zero. Excluding the four eves whose parity is off by more than $0.50 changes the normal-day mean from +0.051% to +0.047% and nothing else.

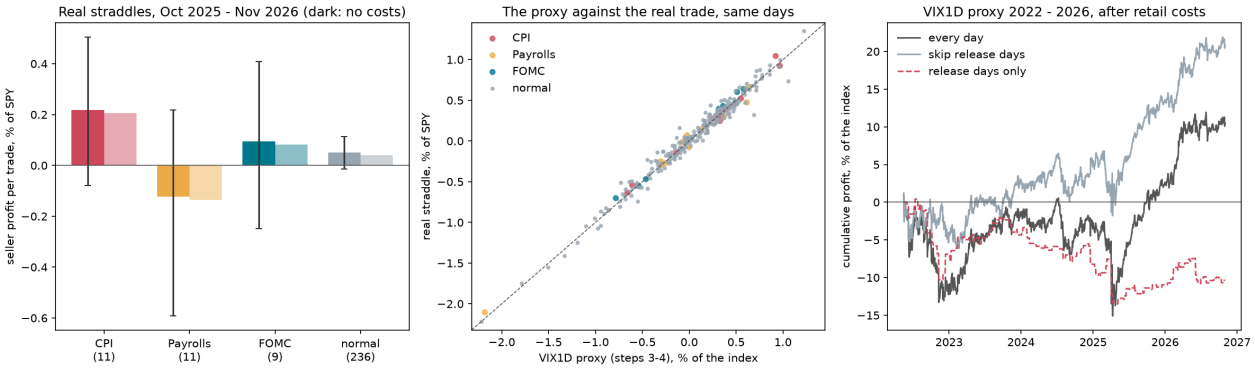


Figure 4. Left: real straddles, seller profit per trade before costs (dark, with 95% CI) and after retail costs (light). Centre: the VIX1D proxy against the real trade on the same days. Right: cumulative profit of the three strategies over 2022–2026, after retail costs.

7. A pricer from scratch

Measuring when event risk leaves option prices (Section 8) requires implied volatility minute by minute for every expiry day, but the vendor supplies it only for release days. A pricer was therefore written for this project. It computes Black–Scholes prices and Greeks (Black and Scholes, 1973; Merton, 1973), solves for implied volatility by bisection, which always converges because the price of an option rises with volatility, and prices American options on a binomial tree (Cox, Ross and Rubinstein, 1979). Its unit tests check it against textbook examples (Hull), put–call parity, finite-difference Greeks and a round trip from volatility to price and back.

Against the vendor. From the same price, SPY price, time and rate, the pricer reproduces the vendor's implied volatility on 46,816 option-minutes to a median of 5×10^{-7} volatility points (largest gap 3×10^{-5}) and its Greeks to a relative 10^{-7}: the vendor uses exactly this model, with calendar time and a 365-day year. On the 2,457 options of the chain snapshot traded at the vendor's timestamp, every implied volatility is within the vendor's rounding of 0.01 points, provided the clock knows that 27 November 2026, the day after Thanksgiving, closes at 13:00.

Early exercise. SPY options are American. On the same tree, the right to exercise early is worth 0.4 cents per share for a one-day at-the-money put, under 0.1% of the straddle, so the European formula is accurate for this study. It matters for deep in-the-money puts, which gain about a day of interest on the strike by exercising: 8 cents for a one-day put 5% in the money, $1.07 at 44 days.

The smile and a rebuilt VIX. On the snapshot, protection against a fall is expensive: for options expiring in 9 days, a put 10% below SPY is priced at 35% volatility and the at-the-money option at 19%. Applying Cboe's VIX formula to SPY options, with prices repriced from the pricer's volatilities at a common SPY price, gives 20.2 for 30 days against an official VIX of 20.8, and 20.9 for 9 days against a VIX9D of 22.0. Both stand about 2 to 3.4 points above at-the-money volatility, because the VIX prices variance across all strikes, crash protection included. VIX1D, used in Sections 4–5, measures the same thing over one day, which is one reason the straddle costs only 0.94 of the move it implies.

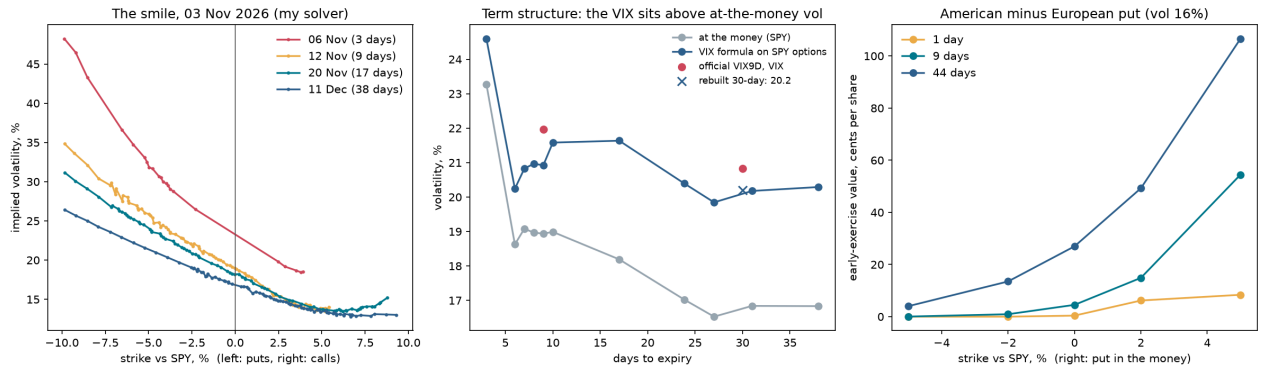


Figure 5. Left: implied volatility across strikes for four expiries on 3 November 2026 (out-of-the-money options, the pricer's volatilities). Centre: the term structure of at-the-money volatility and of the VIX formula applied to SPY options, with the official VIX9D and VIX. Right: value of early exercise for a put, by moneyness and maturity.

8. When does event risk leave the price?

An option expiring at the close of day E prices the variance of SPY's return until that close, $W = \sigma_{impl}^2 \times T$, where T is the time to expiry in years. W falls as the day passes, and falls faster when a risk is removed by an announcement. For each of the 211 expiry days since October 2025 with a one-night eve, W is computed each minute from the 1-minute candles of the eve's at-the-money strike, using the out-of-the-money option of the pair, whose price is all time value. On the release days this matches the vendor's numbers to a median 1.4%.

Overnight. On a normal night, 36% of the eve's W is gone by 09:35 (Table 7). On CPI nights 47% is gone: the release removes 11 percentage points more, a move of about 0.24% that the options priced for CPI itself. On payrolls nights 37% is gone, almost exactly a normal night: **the options price payrolls as barely an event**. Yet

the overnight move on payrolls days was 1.5 times what the options priced for the night [0.99, 1.93], and on CPI days 0.73 times. This is the mechanism behind the payrolls result of Sections 4–6.

Table 7. Priced variance left and actual over priced moves (95% CI), 211 expiry days.

Day	Days	W left at 09:35	Overnight: actual / priced	W left at 15:45 (from 13:55)	Afternoon: actual / priced
CPI	11	0.53	0.73 [0.38, 1.15]	0.58	0.92 [0.42, 1.48]
Payrolls	11	0.63	**1.50 [0.99, 1.93]**	0.64	0.87 [0.41, 1.40]
FOMC	9	0.81	0.62 [0.47, 0.77]	**0.41**	1.17 [0.57, 1.78]
Normal	180	0.64	1.04 [0.92, 1.18]	0.58	0.78 [0.68, 0.89]

W left: median share of the eve's (or 13:55) priced variance still there. Priced move: 0.80 × √(share of W gone, median for the kind of day × W at the start), compared with the actual absolute move over the same interval.

The FOMC afternoon. Relative to 13:55, five minutes after the 14:00 statement 89% of the priced variance is still there, at 14:35 76%, and at 15:00, half an hour into the press conference, 49%, against 73% on normal days at the same time (Figure 6). Most of the risk the options priced for the Fed leaves while the chair is speaking, not when the decision is published. The afternoon move on FOMC days was 1.17 times what was priced, a ratio with a wide interval. On normal days the afternoon is overpriced (0.78) while the night is fairly priced (1.04): the premium of normal days is earned during trading hours.

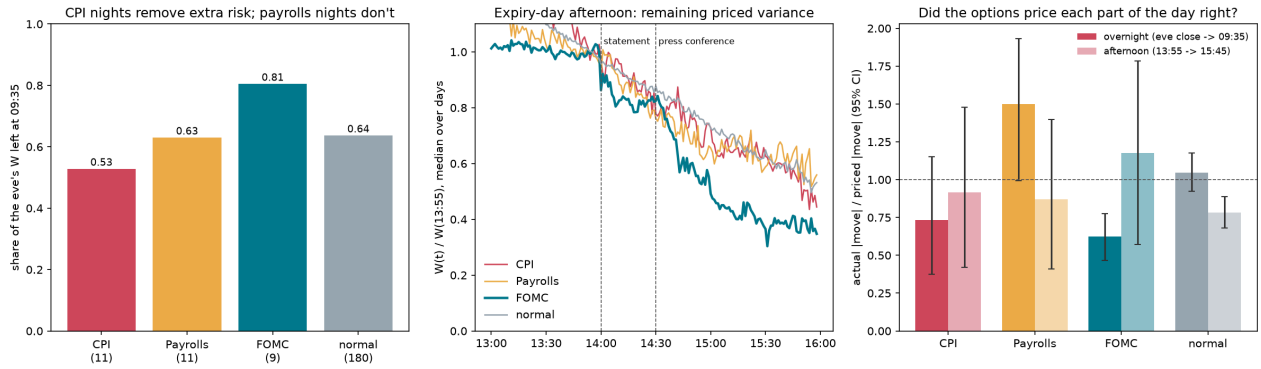


Figure 6. Left: median share of the eve's priced variance left at 09:35. Centre: priced variance through the expiry-day afternoon relative to 13:55, median over days; dashed lines at the 14:00 statement and 14:30 press conference. Right: actual over priced absolute move overnight and in the afternoon (95% CI).

9. Limitations

- **Small samples on release days.** About 50 per type over 4.5 years and 9–11 with real option prices. Every estimate carries a confidence interval, and the central contrast between release and normal days is a consistent pattern, not a significant one.
- **A proxy for most of the sample.** The 2022–26 results use VIX1D, built from index options and all strikes. The 0.94 factor that turns it into a straddle price is measured over one year and ranges from 0.83 to 1.08; the seller's absolute profit depends on it, while comparisons between kinds of day do not. The real trade confirms the proxy where both exist (correlation 0.989).
- **Trades, not quotes.** Costs are scenarios. Market makers who earn the spread do better than Table 6; wider spreads on volatile days would do worse.
- **Tail risk and capital.** Selling straddles loses heavily on rare days. Sharpe ratios do not show the margin a seller needs to survive a day like 9 April 2025.
- **A split chosen after looking.** The CPI regime change is a hypothesis for new data.

10. Conclusion

Options overprice ordinary days, and the premium is robust: on 985 normal days since 2022 the S&P 500 moved 11% less than options had priced. On the days investors fear most, the premium shrinks. Options do charge more ahead of CPI, payrolls and Fed decisions, but not enough to cover the larger moves, so a seller earns nothing on those days on average and a strategy that simply avoids them doubles its risk-adjusted return. The averages hide clear differences between events. Payrolls is the event options miss: they barely price it as an event and it moves the market more than priced. CPI options overcorrected after 2022. And for the Fed, the market fears the press conference more than the statement.

The natural extensions are more data and better prices: quotes rather than trades for costs, real SPY options before October 2025 for a longer test of the payrolls result, and the 2027 releases to test whether CPI options remain expensive.

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Appendix: reproducibility

The full pipeline is public at github.com/unaifmm/options-event-risk. Numbered Python scripts run in order, from the data download (`01_download_data.py`) to the minute-by-minute analysis (`07_intraday.py`), and `python run_all.py` reproduces every figure and table in this paper in about three minutes once the data are downloaded. Twenty-six unit tests check the building blocks (the pricer against textbook values, the settlement calendar, the market clock, the statistics) and run automatically on GitHub at every change. Cboe and FRED data are public; London Strategic Edge data need an account. Raw data are not included. The Python code was written with the help of AI coding assistants; the research questions, design and interpretation are the author's, and every result was checked against the data before it was used. Data downloaded on 4 November 2026. This paper is for research and educational purposes only and is not investment advice.